

University of Engineering and Technology, Lahore

Department of Applied Computing

SOFTWARE REQUIREMENTS SPECIFICATION

(SRS)

QueueSmart – Digital Clinic Queue Management System

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QueueSmart – Digital Clinic Queue Management System

1. Purpose & Problem Statement

Patients in healthcare facilities often face long wait times in crowded waiting rooms. This leads to frustration, wasted time, and in some cases, health risks. QueueSmart aims to replace traditional paper-based token systems with a centralized digital solution. Patients can book virtual tokens, track their real-time position in the queue, and receive automated notifications when their turn is near. The system helps clinics run more smoothly and improves the overall patient experience.

2. Users

Patients: Individuals who book virtual tokens and track their queue status through a mobile app or web portal.

Doctors: Medical practitioners who view the active queue and manage patient flow.

Receptionists / Clinic Staff: Administrative staff who handle walk in patients and operate the central dashboard.

3. Stakeholders

- Patients and their accompanying family members
- Clinic owners, doctors, and medical staff
- Healthcare regulatory bodies interested in patient flow efficiency
- Software development and maintenance team

4. Functional Requirements

Virtual Token Booking: The system shall allow patients to book a time slot or join a live digital queue remotely.

Real-Time Queue Tracking: The system shall display the current token number being served and the patient's estimated position in the queue.

Automated Notifications: The system shall send an SMS or push notification to the patient when they are three tokens away from their turn.

Doctor's Dashboard: The system shall provide an interface for the doctor or receptionist to "Call Next," "Skip," or "Mark Complete" for each patient.

Walk-in Management: The system shall allow the receptionist to generate a digital token for patients who arrive at the clinic without a prior booking.

5. Non-Functional Requirements

Performance: Queue updates and notifications must be delivered in real time with latency under 2 seconds.

Usability: The patient interface must be simple and clear, with large buttons and easy-to-follow instructions suitable for users of all ages.

Security: Patient personal data (names, phone numbers, and medical history) must be encrypted and protected from unauthorized access.

Availability: The system must maintain 99% uptime during standard clinic operating hours.

Scalability: The backend must support multiple clinics and hundreds of concurrent users without performance loss.

6. Risk Assessment

Patient No-Shows: A patient books a token but does not arrive, which can disrupt the queue.

Mitigation: Apply a policy that temporarily restricts priority bookings for patients with repeated no-shows.

Network / Internet Outages: Loss of internet connectivity at the clinic can stop the digital queue.

Mitigation: Provide an Offline Mode on the doctor's dashboard that stores actions locally and syncs them automatically when the connection returns.

Data Privacy Breach: Unauthorized access to patient visit history.

Mitigation: Enforce strict Role-Based Access Control (RBAC) and encrypt sensitive data both at rest and in transit.

7. Quality Attributes

Reliability: Notification and queue-update features must work consistently, as they are central to the system's purpose.

Maintainability: The codebase should be modular and well documented so that future developers can add new features (for example, telemedicine modules) with ease.

Efficiency: The mobile application must use minimal battery and data on the patient's device.

8. AI Integration Strategy

Where AI is used: Predictive Wait Time Estimation. An AI model reviews past consultation data (average time per patient, type of visit, and similar factors) to give more accurate wait-time estimates instead of fixed averages.

What could go wrong: Unexpected delays (such as a medical emergency) can make the AI prediction inaccurate and frustrate patients if they arrive too late or too early.

Mitigation: The system will show a time range (for example, 15 to 25 minutes) rather than a single exact time. Receptionists can also manually override or pause the AI predictions when unusual situations occur.

9. SDLC Implementation Steps

QueueSmart will be developed following the standard Software Development Life Cycle (SDLC) to keep the process organized and professional:

Phase 1 – Requirements Gathering: Conduct structured interviews with clinic staff and patients to record clear functional and non-functional needs.

Phase 2 – System Design: Create UI/UX wireframes for the mobile and web interfaces, and design a secure, normalized database schema.

Phase 3 – Implementation (Coding): Build the frontend with a cross-platform framework (such as Flutter) and the backend with a reliable framework (such as Node.js or Django). Integrate SMS APIs for notifications.

Phase 4 – Testing: Carry out Unit Testing, Integration Testing, and User Acceptance Testing (UAT) in a real clinic setting to find and fix issues.

Phase 5 – Deployment: Host the web dashboard on a secure cloud server (for example, AWS) and publish the mobile application on the Google Play Store.

Phase 6 – Maintenance: Provide ongoing monitoring, release regular updates for OS compatibility, and add improvements based on user feedback.

10. Conclusion

QueueSmart is a practical and scalable digital queue management system designed for healthcare facilities. By following sound software engineering practices from clear requirements analysis through systematic SDLC stages and careful AI use the project shows how technology can reduce crowding, improve clinic operations, and give patients a better experience. The system solves the immediate problem of long physical queues while providing a secure and maintainable base for future digital health features.